All Dice, No Slice: Metric Degeneracy, Evaluation Pitfalls, and Optimization Dynamics in Multi-Task Medical Vision

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Abstract

Hard parameter-sharing U-Nets couple classification and lesion segmentation in resource-constrained medical vision. Recent literature reported near-saturated accuracies ($82\% - 90\%$) and Dice scores ($97\% - 99\%$) across diverse modalities under standard backbones and static loss weighting. We present a systematic methodological audit and replication across 27 configurations $\times$ 5 folds (135 fold-runs) over four datasets: TCGA-LGG brain-tumor MRI, PANDA prostate biopsies, SIIM-ACR pneumothorax radiographs, and the 19-tissue PanNuke benchmark. Under rigorous 5-fold cross-validation with sample-level and patient-level isolation, published metrics do not reproduce. We investigate three key factors governing this divergence: (1) an empty-mask Dice degeneracy where macroscopic slice averages reward all-empty predictions on negative cases, creating an artificial score floor ($\rho = 77.72\%$ on SIIM) that can conceal severe foreground omission; (2) task-formulation divergence in computational pathology, where the baseline reported 88% accuracy on proxy morphological tissue classes (52.3% non-cancer stroma/background) whereas authentic clinical ISUP grading on standard thumbnails yields $28.21\% - 44.87\%$ top-1 accuracy, reflecting different classification targets; and (3) multi-task optimization dynamics where dynamic gradient balancing (GradNorm) is associated with a substantial drop in

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validation accuracy (38.18% → 28.21%, consistent with destructive gradient interference hypotheses). Furthermore, matched ablations show modest paired classification differences for lateral skip connections (+0.02 to −0.12 pts), while bypassing Macenko stain normalization consistently improves diagnostic discrimination (+3.81 pts PANDA, +2.75 pts PanNuke). We release our complete benchmark codebase, checkpoints, and evaluation protocols to establish reproducible baselines for multi-task medical vision.

Keywords: Medical Image Analysis, Multi-Task Learning, U-Net, Reproducibility, Metric Degeneracy, Histopathology

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1. Introduction

Deep learning systems in digital pathology and radiology are increasingly required to deliver both diagnostic categorization and lesion delineation [1, 2]. Large visual foundation models (e.g., UNI [3], CONCH [4], and Virchow [5]) show that clinically actionable representations require massive pretraining and self-supervised objectives across gigapixel whole-slide images (WSIs). In resource-constrained deployments, hard parameter sharing via an encoder–decoder backbone (a U-Net [6] with dual prediction heads) remains a lightweight alternative that couples both objectives in a single forward pass.

Recently, Rhanoui et al. [7] reported that a standard U-Net with lightweight encoders (VGG16 [8] and MobileNetV2 [9]) trained with static loss weights ($\lambda_{seg} = 5.0, \lambda_{cls} = 1.0$) and a high learning rate ($\eta = 10^{-3}$) simultaneously solves both classification and segmentation across four distinct medical modalities (Brain MRI, Skin Dermoscopy, Prostate Histopathology, and Chest X-Ray), reporting $82\% - 90\%$ classification accuracy and $97\% - 99\%$ Dice similarity coefficients.

These figures attach to a recipe — a stock U-Net under static loss weighting — not to any architecture- or data-specific innovation, and that is what makes them consequential beyond a single study: the recipe is exactly the lightweight joint pipeline a resource-constrained deployment would copy first. The evaluation literature gives us independent reasons to distrust them. Region-overlap scores degenerate when annotations are sparse or absent [10, 11]; patch-level splits of tiled WSIs leak patient identity [12, 13]; and joint optimization is documented to interfere with at least one of its tasks [14, 15]. Section 2 reviews each strand. The audit that follows asks whether this particular, highly visible instance of the recipe survives the scrutiny those

literatures demand.

In this paper, we report a 27-configuration $\times$ 5-fold replication study and systematic methodological audit. Our findings demonstrate that:

1. **Task-Formulation Divergence on Prostate Histopathology:** When evaluated under rigorous 5-fold cross-validation on clinical 6-class ordinal ISUP grading (the diagnostic grading standard of the PANDA prostate biopsy benchmark [16]), standard CNN models on 128×128 thumbnails achieve an empirical range of $28.21\% - 44.87\%$ top-1 accuracy (fold-mean, 95% bootstrap CIs $[24.76, 31.47]$ to $[43.93, 45.76]$) across all 12 configurations (Runs 03–04 baseline: 34.51% and 39.91%). As documented in Table 2 of Rhanoui et al. [7], the published 88% accuracy evaluated proxy morphological tissue categories (Background, Stroma, Benign epithelium, Gleason 3/4/5; where background and stroma alone constituted 52.3% of samples). Because tissue categorization and clinical ISUP grading represent different classification tasks, their accuracies are not directly comparable.
2. **The Empty-Mask Dice Metric Degeneracy:** Scoring empty references against empty predictions as Dice = 1.0 introduces a severe score floor on sparse-lesion cohorts. Our SIIM pneumothorax runs converge to all-empty masks on 19 of 20 folds, realizing the all-empty background floor (77.72% pooled across folds for three configurations; 77.04% for the fourth, matching the negative fraction ρ). While the cause of the published 99% Dice cannot be determined without access to the original predictions and code, our analytical derivation and empirical results prove that macroscopic Dice can conceal severe foreground segmentation failure.
3. **Optimization Dynamics in Joint Training:** Matched factorial controls reveal that dynamic gradient balancing is associated with a marked drop in validation accuracy ($38.18\% \rightarrow 28.21\%$ Acc, $\Delta = -9.97$ percentage points under matched conditions, consistent with destructive gradient interference hypotheses). Furthermore, lateral skip connections exhibit modest paired classification differences (+0.02 percentage points on TCGA, -0.12 on PANDA), and omitting Macenko stain normalization consistently yields higher diagnostic discrimination (+3.81 pts on PANDA, +2.75 pts on PanNuke).

To facilitate complete reproducibility, all training scripts, raw

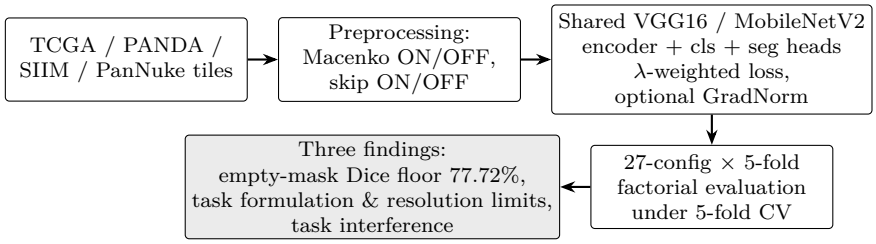


Figure 1: Overview of the 27-configuration × 5-fold replication and audit pipeline. Four source datasets (TCGA-LGG, PANDA, SIIM-ACR, PanNuke) are processed and passed through a modular pipeline toggling Macenko stain normalization and lateral skip connections. A shared VGG16/MobileNetV2 encoder feeds classification and segmentation heads trained with a λ-weighted loss (static ratios or optional GradNorm). All 27 configurations are evaluated under rigorous 5-fold cross-validation (patient-level for TCGA, image-level for PANDA, SIIM, and PanNuke), uncovering three core findings: the empty-mask Dice floor of 77.72% on SIIM (pooled across folds; per-fold pooled 74.33%–77.75%), task-formulation and resolution divergence on PANDA, and task interference from joint multi-task optimization.

JSON summaries for all 27 configurations (135 fold-runs), evaluation checkpoints, and generated figures are publicly available at https://github.com/ApatheticMioz/cancer-pathology-dl [17]. Figure 1 summarizes the end-to-end pipeline and the three findings it yields.

2. Related Work: Documented Failure Modes in Multi-Task Medical Segmentation

None of the three challenges interrogated in this audit is peculiar to the pipeline we examine. Each represents a systemic failure mode that the evaluation, computer vision, and machine-learning literatures have documented independently. This section situates our contribution within those strands; the audit itself instantiates all three within one controlled, artifact-released reproduction of a widely cited joint detection–segmentation recipe.

Metric degeneracy under sparse annotation. Scoring an empty prediction against an empty reference as perfect agreement (Dice = 1.0) is a recognized pitfall in biomedical validation: Reinke et al. [10] catalogue it among the primary metric traps that distort image analysis benchmarks, and Eelbode et al. [11] demonstrate formally that region-overlap scores behave counter-intuitively when annotations are small or absent. Liu et al. [18] show that region-size bias propagates directly into loss formulations. Slice-averaged

Dice on a cohort whose lesion-free fraction approaches 78% sits squarely inside this failure class: Section 3.2 derives the exact decomposition governing what the convention credits.

Multiscale pathology, resolution limits, and evaluation protocols. Clinical computational pathology relies on gigapixel whole-slide images (WSIs) where diagnostic grades emerge from multiscale tissue architecture [2, 16]. Severe spatial downsampling (e.g. compressing biopsy cores to 128×128 thumbnails) obliterates cellular atypia and gland lumen boundaries, turning cells subpixel. Furthermore, patch-level random splits of tiled WSIs risk severe data leakage when tiles from the same patient or biopsy core contaminate both train and test partitions [12, 13]. When evaluating pathology AI, isolating independent sampling units and aligning tasks with clinical diagnostic targets (rather than proxy tissue textures) is essential for clinical validity.

Task interference and negative transfer. Whether classification and segmentation reinforce or degrade one another under a shared encoder depends heavily on task affinity: Standley et al. [14] observe that naive co-training frequently induces negative transfer across dense and global objectives; Crawshaw's survey [15] collects extensive evidence of gradient conflict; and dynamic gradient balancing [19] is often proposed as a remedy. However, dynamic balancing can destabilize optimization when task losses exhibit disparate convergence dynamics: while Kang et al. [20] report benefits for wound-tissue analysis, our experiments demonstrate that GradNorm forces destructive updates on complex ordinal classification heads.

Propagation and citation landscape. These strands converge on the published recipe of Rhanoui et al. [7], which reported $97\% - 99\%$ Dice alongside $82\% - 90\%$ accuracy across four modalities. Because multi-task parameter sharing is widely sought for low-resource clinical deployments, these headline figures have been cited uncritically as evidence of general multi-task efficacy in follow-on oncology work. Notably, recent works such as Al Hassan et al. [21] have specifically identified the need for leakage-safe validation and external cohorts when evaluating joint brain tumor models. Supplying a rigorous, empirical, and mathematical quantification of how such numbers arise is the primary focus of this work.

3. Multi-Task Formulation & Evaluation Metrics

3.1. Architectural Formulation

Let $\mathbf{x} \in \mathbb{R}^{H\times W\times C}$ denote an input image tile. A shared convolutional encoder $f_\theta(\mathbf{x})$ extracts a bottleneck latent representation $\mathbf{z} \in \mathbb{R}^{h\times w\times d}$. The classification branch applies global average pooling (GAP) followed by a two-layer multilayer perceptron (hidden width 256, dropout 0.5):

$$\hat{\mathbf{y}}_{cls} = \mathrm{Softmax}(\mathbf{W}_2\,\phi(\mathbf{W}_1\,\mathrm{GAP}(\mathbf{z}) + \mathbf{b}_1) + \mathbf{b}_2) \in \Delta^{K-1}, \tag{1}$$

where K is the number of diagnostic classes, Δ^{K-1} denotes the probability simplex, and $\phi(\cdot)$ is the ReLU nonlinearity. The segmentation branch passes $\mathbf{z}$ through a decoder Dec_ϕ with lateral skip connections $\{\mathbf{s}_l\}_{l=1}^{L}$ from encoder stages, yielding unnormalized pixel logits $\mathbf{l}_{seg} \in \mathbb{R}^{H\times W\times M}$. For binary segmentation ($M = 1$), $\hat{\mathbf{y}}_{seg} = \sigma(\mathbf{l}_{seg})$ with $\sigma(\cdot)$ the sigmoid; for $M > 1$, class probabilities are obtained via channel-wise Softmax.

The joint multi-task training objective is defined as:

$$\mathcal{L}_{total}(\theta, \phi, \mathbf{W}_{cls}) = \lambda_{cls}\mathcal{L}_{cls}(\mathbf{y}_{cls}, \hat{\mathbf{y}}_{cls}) + \lambda_{seg}\mathcal{L}_{seg}(\mathbf{y}_{seg}, \mathbf{l}_{seg}), \tag{2}$$

where $\mathcal{L}_{cls}$ is categorical cross-entropy $\mathcal{L}_{CE}$. To address class imbalance, the classification loss incorporates inverse-frequency balanced class weights $w_c = N/(K \cdot N_c)$ [22], matching the weighting formulation specified in Rhanoui et al. [7] and implemented in our training pipeline. The segmentation loss is purely cross-entropy without an active Dice-based gradient penalty:

$$\mathcal{L}_{seg}(\mathbf{y}_{seg}, \mathbf{l}_{seg}) = \begin{cases} \mathcal{L}_{BCE}(\mathbf{y}_{seg}, \mathbf{l}_{seg}), & M = 1, \\ \mathcal{L}_{CE}(\mathbf{y}_{seg}, \mathbf{l}_{seg}), & M > 1, \end{cases} \tag{3}$$

evaluating binary cross-entropy on binary tasks (TCGA, SIIM) and multi-class cross-entropy on multi-class tasks (PANDA, PanNuke). Rhanoui et al. formulated segmentation as a binary cancer-vs-background detection task across all datasets; our audit matches this binary objective for TCGA and SIIM, while evaluating multi-class cross-entropy for PANDA and PanNuke to directly benchmark their multi-category annotations. Both pipelines incorporate standard spatial data augmentations (horizontal and vertical flips, random rotation $\pm 15°$, and affine shear $\pm 10°$).

The Dice similarity coefficient (DSC) is evaluated strictly post-hoc as a validation metric, not as an active loss term during training. While cross-entropy minimization can in principle yield segmentations with high spatial overlap, evaluating Dice post-hoc on sparse and negative-dominant cohorts makes the metric vulnerable to evaluation conventions, as derived below.

3.2. The Empty-Mask Dice Metric Degeneracy

The Sørensen–Dice coefficient between ground-truth mask Y and binary prediction $\hat{Y}$ ($Y, \hat{Y} \in \{0,1\}^{H\times W}$) is:

$$\mathrm{Dice}(Y,\hat{Y}) = \frac{2|Y \cap \hat{Y}|}{|Y| + |\hat{Y}| + \epsilon}, \quad \epsilon = 10^{-8};\ \ \mathrm{Dice} := 1 \text{ if } |Y| + |\hat{Y}| = 0. \tag{4}$$

Empty references and predictions represent a known validation trap in biomedical image analysis [10]. Two common conventions govern empty slices:

- **Foreground-Only Evaluation:** Compute Dice strictly over non-empty target slices ($\{i \mid |Y_i| > 0\}$).
- **Empty-Credit Assignment:** Assign Dice = 1.0 when $|Y| = 0 \wedge |\hat{Y}| = 0$, and Dice = 0.0 when $|Y| = 0 \wedge |\hat{Y}| > 0$.

Averaged across all N slices under empty-credit assignment, the macroscopic mean is:

$$\overline{\mathrm{Dice}}_{\mathrm{all}} = \tfrac{1}{N}\textstyle\sum_{i=1}^{N} d_i, \quad d_i = \mathrm{Dice}(Y_i,\hat{Y}_i) \text{ if } |Y_i| > 0, \text{ else } \mathbf{1}\left[|\hat{Y}_i| = 0\right]. \tag{5}$$

Let $\rho = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[|Y_i| = 0]$ denote the proportion of ground-truth negative (lesion-free) slices, and let $q_{\mathrm{empty}} = P(|\hat{Y}| = 0 \mid |Y| = 0)$ denote the model's true-negative prediction rate on negative slices. The exact macroscopic identity is:

$$\overline{\mathrm{Dice}}_{\mathrm{all}} = \rho \cdot q_{\mathrm{empty}} + (1-\rho)\cdot \overline{\mathrm{Dice}}_{\mathrm{fg}}, \tag{6}$$

where $\overline{\mathrm{Dice}}_{\mathrm{fg}}$ is the mean Dice score over lesion-bearing slices ($|Y_i| > 0$).

When all ground-truth negative images are predicted empty ($q_{\mathrm{empty}} = 1$), the identity simplifies to $\overline{\mathrm{Dice}}_{\mathrm{all}} = \rho + (1-\rho)\cdot \overline{\mathrm{Dice}}_{\mathrm{fg}}$. If a model produces false-positive segmentations on negative images, $q_{\mathrm{empty}} < 1$. For a degenerate, completely collapsed predictor that emits empty masks on all inputs ($|\hat{Y}| = 0$ everywhere), foreground Dice is zero ($\overline{\mathrm{Dice}}_{\mathrm{fg}} = 0$) while $q_{\mathrm{empty}} = 1$, yielding $\overline{\mathrm{Dice}}_{\mathrm{all}} = \rho$.

Consequently, on sparse-lesion cohorts such as SIIM-ACR pneumothorax ($\rho = 77.72\%$), an all-empty baseline scores 77.72% solely from empty-mask credit. Crucially, as shown by Eq. 6, reaching an overall score of 99% on such a cohort requires positive-case foreground Dice of approximately $\overline{\mathrm{Dice}}_{\mathrm{fg}} \approx$

$(0.99 - 0.7772)/(1 - 0.7772) \approx 95.51\%$ under perfect negative predictions ($q_{\text{empty}} = 1$). Our SIIM runs demonstrate how overall Dice can conceal foreground failure: they emit all-empty masks on 19 of 20 folds, directly realizing the all-empty score floor (77.72% pooled across folds for Runs 05, 06, and 11; 77.04% for Run 12 due to a small non-empty residual on fold 4; per-fold range $74.33\% - 77.75\%$). While the definitive cause of the published 99% Dice in Rhanoui et al. cannot be determined without access to their predictions, masks, and scoring scripts, this empirical collapse demonstrates that macroscopic Dice on sparse cohorts can register high scores despite zero foreground delineation. Fig. 2 illustrates this degeneracy.

This metric inflation generalizes to multi-class histopathology. On 6-class PANDA prostate biopsies, individual tiles typically contain only one or two histological patterns, leaving 4–5 classes absent ($|Y_c| = 0$). Under channel-wise empty crediting, an all-background model receives Dice = 1.0 for every absent class alongside high agreement on background. Throughout this paper, we track two distinct estimators:

1. *Training-Time Batch Validation Macro-Dice:* Computed during training validation via batch-pooled channel-wise Dice (`dice_coefficient`), crediting an empty channel across the batch with 1.0, and averaging across all M channels (fold-mean across 12 PANDA configurations: $82.77\% - 84.39\%$).
2. *All-Slices Empty-Credit Pooled Per-Case Dice:* Pools all validation slices across folds, credits empty-over-empty cases with Dice = 1.0, and averages per case (yielding the $98.54\% - 98.60\%$ floor on PANDA, despite near-zero true glandular delineation; Fig. 5).

3.3. Adaptive Gradient Balancing via GradNorm

To test whether dynamic loss balancing mitigates cross-task gradient competition, we introduced a parameterized variant of GradNorm [19] in Phase v2.

Let W denote the shared convolutional filters and normalization layers of the U-Net encoder f_θ. The L_2 gradient norm of task $i \in \{seg, cls\}$ with respect to W at minibatch t is:

$$G_W^{(i)}(t) = \|\nabla_W w_i(t)\mathcal{L}_i(t)\|_2 . \tag{7}$$

For stability in the primary `Adam` optimizer, balancing weights are learnable log-weights $\mu_i(t) = \ln w_i(t)$ (initialized to $w_{seg}(0) = 5.0$, $w_{cls}(0) = 1.0$),

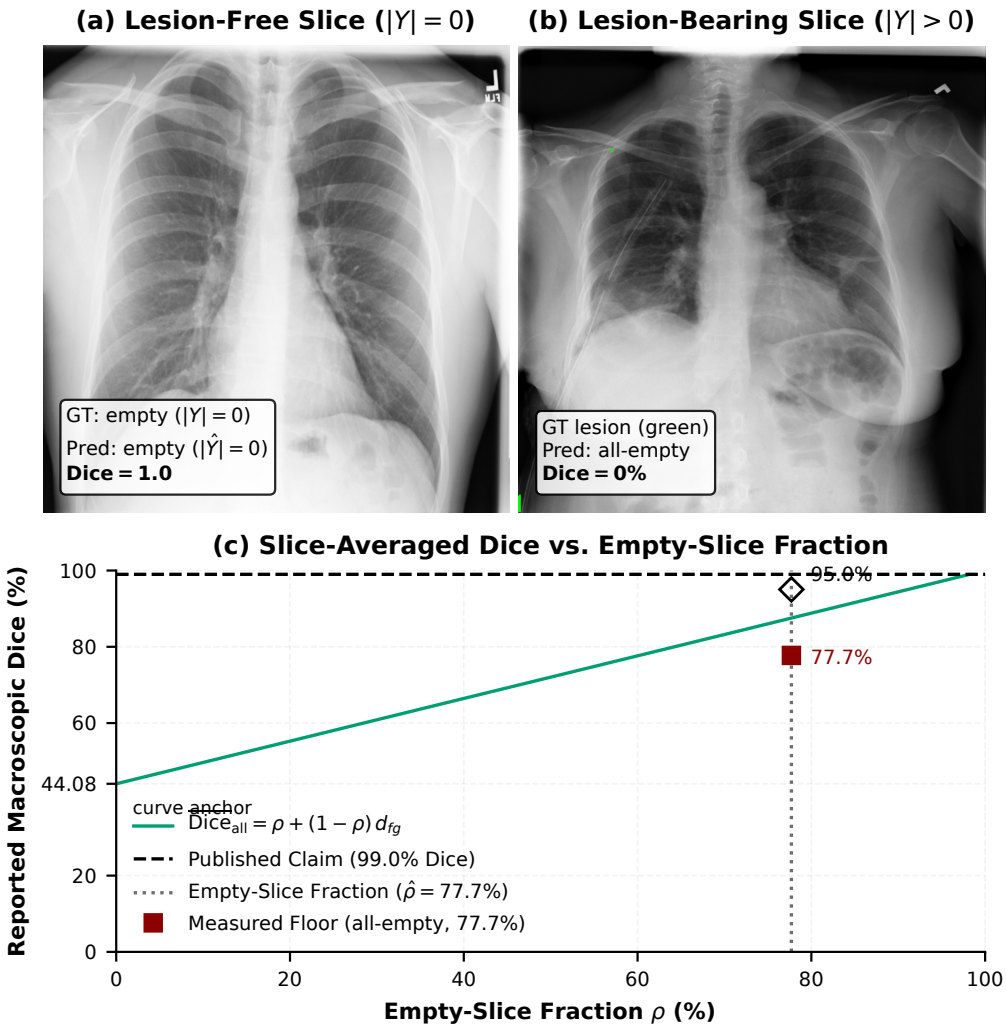


Figure 2: The empty-mask Dice artifact on SIIM-ACR pneumothorax. (a) Lesion-free slice ($|Y| = 0$): under empty-credit assignment an all-empty prediction is awarded Dice = 1.0; lesion-free slices constitute $\hat{\rho} = 77.7\%$ of our validation split. (b) Lesion-bearing slice ($|Y| > 0$, ground-truth boundary in green): our all-empty models predict empty masks here and score Dice = 0, invisible to the slice-averaged mean. (c) Exact macroscopic curve (Eq. 6): a foreground Dice of 77.72% reports 95.03% at $\hat{\rho} = 0.777$ under perfect negative prediction ($q_{\text{empty}} = 1$), whereas our SIIM runs measure the all-empty floor itself (77.72% pooled; 77.04% for the run with one residual fold).

updated jointly with the network parameters. Minibatch optimization minimizes the L_1 gradient disparity objective:

$$\mathcal{L}_{grad}(t) = \sum_{i \in \{seg,cls\}} \left| G_W^{(i)}(t) \; - \; \bar{G}_W(t) \cdot [r_i(t)]^{\alpha} \right|, \tag{8}$$

where $\bar{G}_W(t) = \frac{1}{2} \sum_j G_W^{(j)}(t)$ is the average gradient norm across tasks, $\alpha = 1.5$ is the asymmetry parameter, and $r_i(t)$ represents the relative inverse training rate:

$$r_i(t) = \frac{\tilde{\mathcal{L}}_i(t)}{\frac{1}{2} \sum_{j \in \{seg,cls\}} \tilde{\mathcal{L}}_j(t)}, \quad \text{with} \quad \tilde{\mathcal{L}}_i(t) = \frac{\mathcal{L}_i(t)}{\mathcal{L}_i(0)}, \tag{9}$$

where $\mathcal{L}_i(0)$ is the initial step-0 loss. Following Chen et al. [19], slower-training tasks exhibit higher loss ratios $r_i(t) > 1$, which raises their target gradient norm $\bar{G}_W(t)[r_i(t)]^{\alpha}$. After each optimizer update, we renormalize the weights so that $\sum_i w_i(t) = 2$. In the canonical formulation [19], the module differentiates $\mathcal{L}_{grad}$ against detached targets and applies it to the balancing weights alone through a dedicated optimizer (learning rate 0.025). §5.3 probes this distinction empirically. We evaluate the module across Groups 2 and 4 to isolate how dynamic gradient equalization behaves in hard-parameter-sharing architectures.

3.4. Color Deconvolution and Optical Density Space

To mitigate staining variability across clinical laboratories, the v2 pipeline incorporates Macenko stain normalization [23]. We convert RGB pixel intensities $\mathbf{I} \in [0, 255]^3$ to optical density (OD) space:

$$\mathbf{OD} = -\ln\left(\frac{\mathbf{I} + \epsilon}{I_0}\right), \tag{10}$$

where $I_0 = 255$ represents transmitted light intensity. The implementation computes optical density via natural logarithms $-\ln(\cdot)$ rather than the canonical base-10 formulation $-\log_{10}(\cdot)$ [23]. We estimate the hematoxylin and eosin stain vectors by eigendecomposition of the optical-density covariance matrix (stain-vector angles restricted to the 1st–99th percentile range) over OD values above a threshold of 0.15, and project them onto a target reference vector. This step effectively eliminates macroscopic batch variations. However, projecting raw RGB channels onto static stain vectors alters chromatic contrast and stain concentration profiles across the tile, potentially attenuating subtle chromatin textures and colour variations that contribute to distinguishing fine glandular structures (Fig. 3).

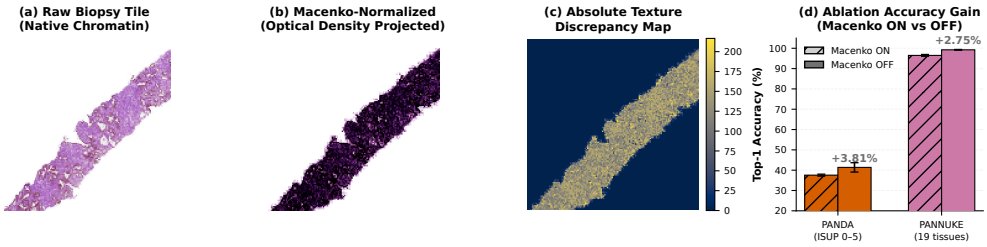


Figure 3: Visual and quantitative impact of Macenko stain normalization on histopathology classification. (a) Raw prostate biopsy tile. (b) Optical-density-projected tile: color deconvolution alters chromatic dynamic range and stain concentration profiles across the tile. (c) Absolute texture discrepancy map. (d) Empirical top-1 accuracy gains across matched models when Macenko normalization is bypassed (+3.81% on PANDA, +2.75% on PanNuke).

Table 1: Per-dataset summary of the 27-configuration × 5-fold reproduction campaign (26 numbered runs plus the canonical decoupled GradNorm probe). Val Acc reports 5-fold mean accuracy; Val Dice reports across-folds per-case bootstrap Dice (with per-epoch validation macro-Dice discussed in Section 3.2 and Section 5.2). Ranges span configurations within each dataset under 5-fold group-aware cross-validation. Individual run values with 95% bootstrap CIs are tabulated in `paper_results_matrix_with_ci.csv` and plotted in Figure 4. The SIIM-ACR Dice (77.72% pooled across folds for three configurations and 77.04% for the fourth) and PANDA all-slices empty-credit Dice (98.54%–98.60%) reflect the empty-mask degeneracy floor (Section 3.2); PANDA per-epoch validation macro-Dice is 82.77%–84.39%. †Non-comparable task targets: Rhanoui et al. evaluated 6-class morphological tissue categorization (Table 2 of [7]: Background, Stroma, Benign epithelium, Gleason 3, 4, 5; 52.3% background/stroma) and binary cancer-vs-background segmentation, whereas our audit evaluates clinical 6-class ordinal ISUP grading (grades 0–5) and 6-class multi-category gland segmentation; these metrics evaluate different targets and are not directly comparable.

Dataset	Runs n	Val Acc (%)	Val Dice (%)	ΔAcc (pts)	Pub. Acc (%)	Pub. Dice (%)
TCGA-LGG	5	92.86–95.62	85.33–88.85	2.86–6.62	89.00–90.00	97.00–98.00
PANDA†	12	28.21–44.87	98.54–98.60	−58.79–−42.13†	87.00–88.00†	98.00–99.00†
SIIM-ACR	4	77.71–82.63	77.04–77.72	−4.81–0.57	82.00–87.00	99.00
PanNuke	5	78.70–99.24	68.44–77.61	—	—	—

4. The 27-Configuration × 5-Fold Empirical Reproduction Campaign

4.1. Datasets and Experimental Protocol

We evaluated four benchmark datasets spanning diverse anatomical modalities:

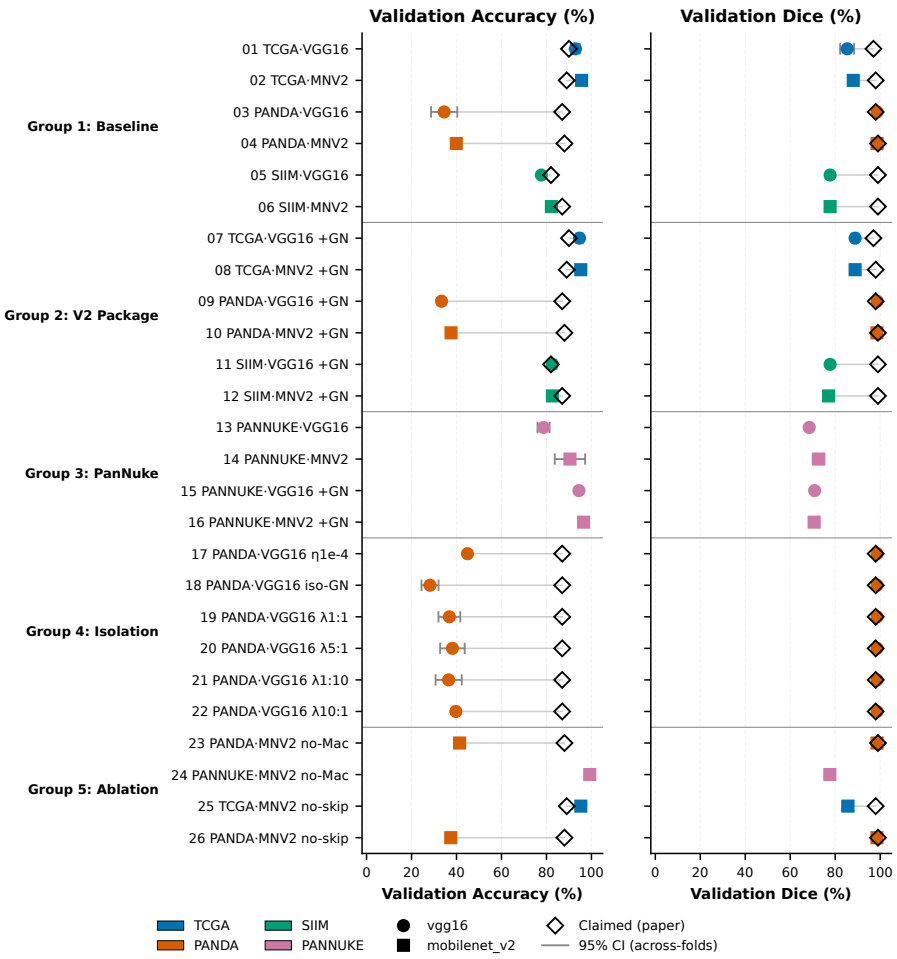


Figure 4: Claimed-vs-measured dot plot for the 26 numbered runs of the 27-configuration × 5-fold campaign (the canonical decoupled GradNorm arm, the campaign's 27th configuration, is overlaid in Fig. 6). Left panel: fold-mean validation accuracy (%) with 95% bootstrap-CI whiskers (measured, filled markers) and the published claimed accuracy (black open diamond); tie lines connect measured to claimed values. Right panel: all-slices empty-credit per-case Dice (%) with 95% across-folds bootstrap CIs (published Dice values have no confidence intervals and are shown as diamonds). For PANDA (Runs 03–04, 09–10, 17–22, 23, 26), measured accuracy spans 28.21%–44.87% vs. published 87%–88%, and all-slices empty-credit Dice spans 98.54%–98.60% (per-epoch validation macro-Dice 82.77%–84.39%). Crucially, as documented in Rhanoui et al. Table 2, published PANDA accuracy evaluated 6 proxy morphological tissue categories rather than the clinical ordinal ISUP grading evaluated here; their accuracies are not directly comparable. All four SIIM rows (Runs 05, 06, 11, 12) measure the all-empty floor (77.72% pooled across folds for three configurations, 77.04% for the fourth; per-fold pooled 74.33%–77.75%). PanNuke rows (Runs 13–16, 24) have no published targets and are shown without claimed diamonds.

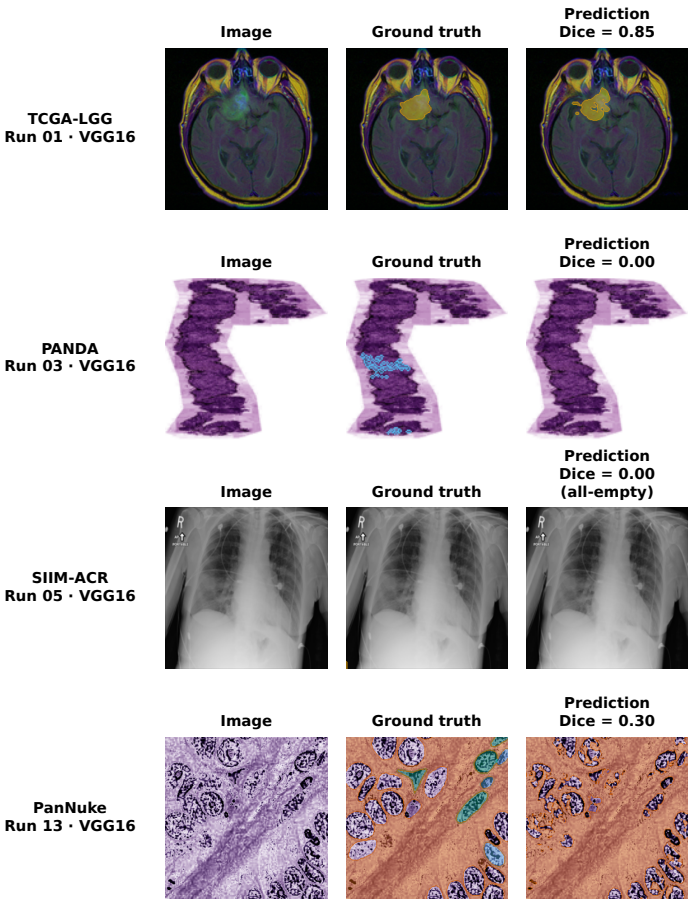


Figure 5: Qualitative ground-truth vs. prediction validation tiles for the four canonical v1 naked VGG16 baselines (Runs 01, 03, 05, 13; their best checkpoints). Each row is one dataset (TCGA-LGG, PANDA, SIIM-ACR, PanNuke); the three columns are the input image, the ground-truth mask overlay, and the model's prediction overlay, with the per-tile Dice (binary Dice for TCGA/SIIM; macro Dice over the classes present in the ground truth for PANDA/PanNuke) shown in the prediction panel. The validation split is one of the five cross-validation folds (the fold shown here is the first). Tile selection is deterministic: the first validation tile (index order) with a non-empty ground-truth mask; for SIIM-ACR the selection advances to the first lesion-bearing tile whose prediction is all-empty, making the empty-mask degeneracy of Section 3.2 the visible face of the figure.

1. **TCGA-LGG Brain Tumor:** 2-class lesion classification (no tumor vs. tumor) and FLAIR abnormality segmentation on 2D 256×256 MRI slices ($N = 3,929$; 110 patients [24]; strictly partitioned via patient-level `GroupKFold` isolation [22]).
2. **PANDA Prostate Biopsies:** Clinical 6-class ordinal ISUP Gleason grading (grades 0–5) and gland segmentation ($N = 10,516$ tiles [16]; 5-fold cross-validation). In the baseline study of Rhanoui et al. [7], the reported 6 classes were morphological tissue categories from Radboud segmentation masks (Background, Stroma, Benign epithelium, Gleason 3, 4, 5; Table 2 of [7]), where non-cancerous background and stroma alone constituted 52.3% of the dataset ($5,558$ of $10,618$ images). Moreover, Karolinska biopsies in PANDA lack these granular categories. Our experiments evaluate clinical 6-class ordinal ISUP grading; because these represent distinct classification targets, their accuracies are not directly comparable. Under the tested CNN architectures on 128×128 thumbnails, clinical ISUP grading achieves $28.21\% - 44.87\%$ top-1 accuracy.
3. **SIIM-ACR Pneumothorax:** Binary chest-radiography detection and pleural boundary segmentation ($N = 10,675$ images [25]; because the public release contains exactly one image per patient, sample-level grouping coincides with patient-level grouping, yielding a stratified 5-fold partition over image-level labels).
4. **PanNuke (Multi-Organ Benchmark):** 19-tissue multi-organ histopathology benchmark ($N = 7,901$ patches [26]; image-level 5-fold split). The baseline study by Rhanoui et al. reported joint multi-task performance across dermoscopic images from ISIC 2018 [27]. However, in the official ISIC 2018 release, Task 1 provided lesion segmentation masks for $2,594$ images, whereas Task 3 provided disease classification labels for $10,015$ images. Because Rhanoui et al. released no split manifests, code, or explanation of how joint multi-task training was conducted across cohorts with differing annotation availability (e.g., whether unannotated images were masked or omitted), their experimental protocol cannot be reconstructed. We therefore evaluated PanNuke, which provides fully paired, verified 19-class tissue labels and 6-class cell segmentation masks across diverse organs.

We trained all models with PyTorch (up to 50 epochs with early stopping, patience 10; batch size 32; fixed seed 42 per fold) using ImageNet-

pretrained encoders and the `Adam` optimizer with a constant per-phase learning rate and zero weight decay, and we applied no learning-rate schedule or additional regularization [28]. Training ran under bf16 autocast AMP with PyTorch's `GradScaler` [29], with the scaler's `unscale_()` called before gradient-norm clipping (max-norm 1.0) at every optimizer step [30, 31], and the GradNorm log-weights bounded to $[\pm \log 10]$ (capping each task weight at 10). The true pre-clip gradient norm is logged per epoch, and clipping engagement is dataset-dependent: final-10-epoch per-fold mean engagement rates are 0.285 (TCGA, VGG16), 0.516 (PanNuke, no Macenko), 0.879 (PanNuke, VGG16), 0.943 (SIIM, VGG16), and 0.999 (PANDA) — on three datasets (PanNuke VGG16, SIIM, PANDA) clipping is an active, frequently-binding constraint, not a dormant safeguard. Non-finite gradients triggered 52 bounded single-batch skips across the 135 fold-cells, overwhelmingly isolated events (the 25-skip-per-fold cap was never reached); each skipped batch is excluded from epoch metrics and stamped in the fold summaries. One fold (PanNuke, VGG16, final, fold 1) hit a non-finite forward loss at epoch 46; the guard aborted it loudly, and training resumed from the epoch-45 checkpoint, completing epochs 46–50 with zero skips — a transient bf16 overflow, disclosed here as a resumed fold. `torch.compile` (`max-autotune`) was used only in the Phase v1 replication arms by design; Phase v2 arms run eager, and one Phase v1 fold degraded to eager after a Triton capture failure (logged). Mid-campaign the resume/identity machinery was hardened (checkpoint fingerprints, atomic state writes); all 135 fold-cells of record were produced under the fixed regime and re-verified from the fold summaries. In Phase v1 (Runs 01–06), we replicated the original static setup ($\eta = 10^{-3}$, $\lambda_{seg} : \lambda_{cls} = 5 : 1$, raw inputs). In Phase v2 (Runs 07–12), we incorporated dynamic GradNorm ($\alpha = 1.5$, $\eta = 10^{-4}$) and Macenko stain normalization [23]. Groups 3–5 (Runs 13–26) systematically evaluate multi-organ generalizability, optimization hyperparameter isolation, and architectural skip ablations. Accuracy is reported as the five-fold mean; 95% CIs are fold-level bootstrap percentiles (10,000 resamples of the five fold accuracies with replacement, seed 42), and Dice CIs are the across-folds per-case bootstrap intervals tabulated in the released matrix. Table 1 summarizes the campaign per dataset and Figure 4 plots every run.

5. Discussion & Methodological Anatomy

5.1. PANDA Task-Formulation Divergence and Optimization Dynamics

The largest numerical divergence in Table 1 sits on the PANDA prostate biopsy benchmark. The original authors reported 88.0% classification accuracy alongside a 99.0% Dice score. Under our 12 PANDA configurations, clinical 6-class ordinal ISUP grading yields a fold-mean top-1 accuracy spanning $28.21\% - 44.87\%$ (95% bootstrap CIs $[24.76, 31.47]$ to $[43.93, 45.76]$), while 6-class segmentation yields a mean validation macro-Dice of $82.77\% - 84.39\%$ (and an all-slices empty-credit Dice floor of $98.54\% - 98.60\%$). The static 5:1-weighted 10^{-3} baselines (Runs 03–04) reach 34.51% and 39.91% accuracy; Run 17 (the isolated-learning-rate arm at $\eta = 10^{-4}$, 44.87%) represents the strongest configuration in our campaign.

This performance divergence reflects fundamentally distinct task formulations for both objectives. For classification, as documented in Table 2 of Rhanoui et al. [7], the cited study evaluated 6 morphological tissue classes extracted from Radboud segmentation masks (Background, Stroma, Benign epithelium, Gleason 3, 4, 5), where non-cancerous background glass and benign stroma comprised over 52% of the samples ($5,558$ of $10,618$ images). In contrast, our audit evaluates authentic clinical 6-class ordinal ISUP grading (grades 0–5), which requires resolving subtle architectural glandular patterns across the biopsy core [16]. The cited study's 88% accuracy and our $28.21\% - 44.87\%$ accuracy evaluate different classification tasks and are therefore not directly comparable. For tested 2D CNN models on 128×128 thumbnails, clinical ISUP grading achieves $28.21\% - 44.87\%$, highlighting the challenge of ordinal grading under low-resolution inputs without multi-scale attention or multiple-instance learning [16]. Similarly, for segmentation, Rhanoui et al. formulated a binary cancer-vs-background objective (optimizing binary cross-entropy, Section 4.2 of [7]), whereas our PANDA audit benchmarks 6-class multi-category gland segmentation across the full annotation set (yielding $82.77\% - 84.39\%$ mean fold validation macro-Dice, and an all-slices empty-credit Dice floor of $98.54\% - 98.60\%$). Because the targets and scoring conventions differ, the segmentation values are not directly comparable without aligning label spaces. Figure 6 traces the training dynamics and isolates the optimization mechanisms involved.

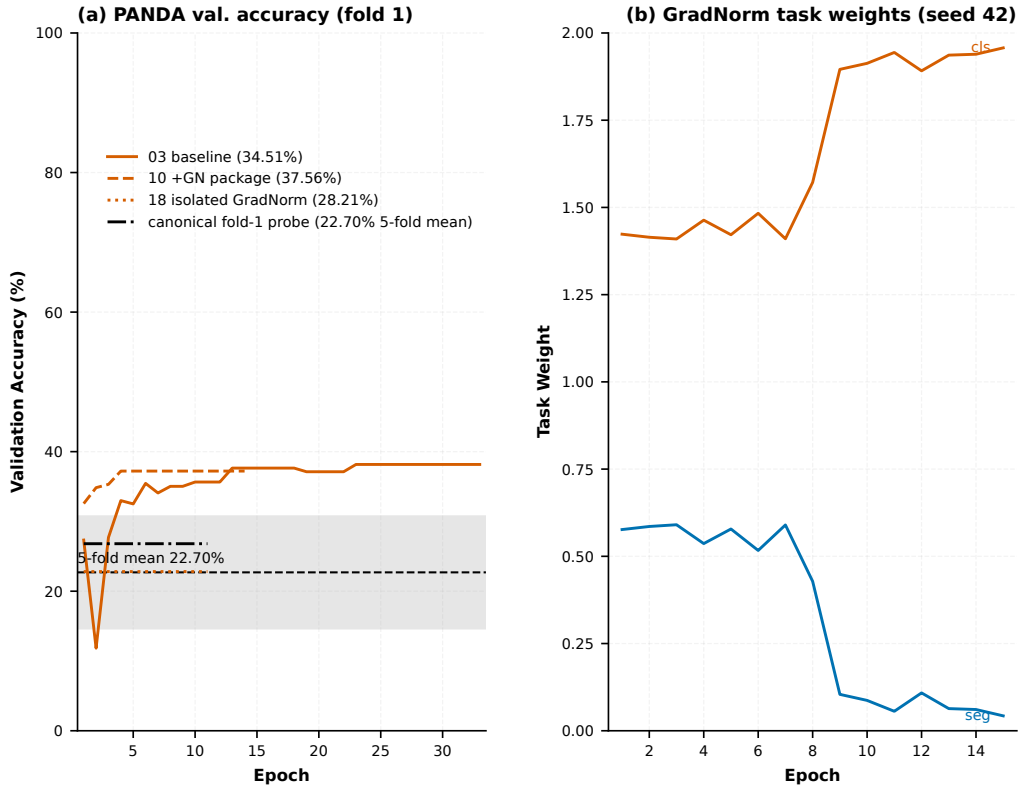


Figure 6: Anatomy of PANDA clinical ISUP grading dynamics. (a) Validation accuracy vs. epoch (fold 1 of 5) for three PANDA runs evaluating clinical ISUP grading — Run 03 (static 5:1 baseline), Run 10 (v2 GradNorm package), Run 18 (isolated GradNorm at $\eta = 10^{-3}$) — with legend values giving each run's five-fold mean accuracy (34.51%, 37.56%, 28.21%); the canonical decoupled GradNorm arm is overlaid (dash-dot, fold-1 trajectory), with its five-fold mean 22.70% (horizontal line) and 95% bootstrap CI $[14.66, 30.73]$ (shaded band) marked. As noted in Section 5.1, these curves illustrate ordinal ISUP grading, which is a different classification objective from the proxy tissue classes reported in Rhanoui et al. (b) GradNorm task-weight dynamics as recorded by the dedicated canonical optimizer probe (15 epochs; per-epoch weights are logged only by that probe): the segmentation weight collapses while the classification weight grows, showing the dynamic rebalancing that the joint variant turns into destructive updates on the shared encoder.

5.2. Controlled Skip Connection Ablation

We evaluate lateral skip connections in matched pairs, holding learning rate, GradNorm, and Macenko normalization identical:

- **Coarse macro-tumor structures (TCGA):** Run 25 (skip-ablated) vs. matched baseline Run 08 (both $\eta = 10^{-4}$, GradNorm, Macenko): accuracy 95.28% vs. 95.26% ($\Delta_{\text{Acc}} = +0.02$ percentage points), Dice 86.16% vs 89.22% ($\Delta_{\text{Dice}} = -3.06$ in mean fold validation macro-Dice; across-folds per-case bootstrap Dice: 85.42% [85.11, 85.73] vs. 88.85% [88.58, 89.12]).
- **Fine micro-glandular structures (PANDA):** Run 26 (skip-ablated) vs. matched baseline Run 10: accuracy 37.44% vs. 37.56% ($\Delta_{\text{Acc}} = -0.12$ percentage points), Dice 84.29% vs 82.77% ($\Delta_{\text{Dice}} = +1.52$ in mean fold validation macro-Dice; across-folds per-case bootstrap Dice: 98.54% [98.36, 98.68] vs. 98.54% [98.38, 98.67]).

The observed classification differences are +0.02 percentage points on TCGA and −0.12 on PANDA, and the marginal 95% bootstrap CIs across the five folds overlap substantially (TCGA: $[94.14, 96.28]$ vs $[94.24, 96.39]$; PANDA: $[36.73, 37.99]$ vs $[37.20, 37.94]$). While this descriptive interval overlap indicates no large separation between the paired configurations, this analysis does not establish equivalence or formally test the significance of the paired differences. The fold-level bootstrap characterizes variability across the five cross-validation partitions rather than uncertainty over repeated independent training campaigns.

Macenko stain-normalization ablation.. The matched Macenko pair (Group 5, Runs 23–24) exhibits an observable classification shift under matched settings. Omitting normalization raises PANDA top-1 accuracy by +3.81 percentage points (Run 23: 41.38% vs. Run 10: 37.56%) and PanNuke accuracy by +2.75 percentage points (Run 24: 99.24% vs. Run 16: 96.49%) with non-overlapping marginal 95% bootstrap CIs across the five folds ($[39.76, 43.38]$ vs $[37.20, 37.94]$; $[99.11, 99.37]$ vs $[96.18, 96.90]$). Segmentation Dice also shifts: on PANDA, mean fold validation macro-Dice is 84.39% vs. 82.77% ($\Delta = +1.62$; across-folds per-case bootstrap Dice: 98.54% vs. 98.54%); on PanNuke, mean fold validation macro-Dice is 70.76% vs. 62.25% ($\Delta = +8.51$; across-folds per-case bootstrap Dice: 77.61% [77.10, 78.12] vs. 70.64% [69.96,

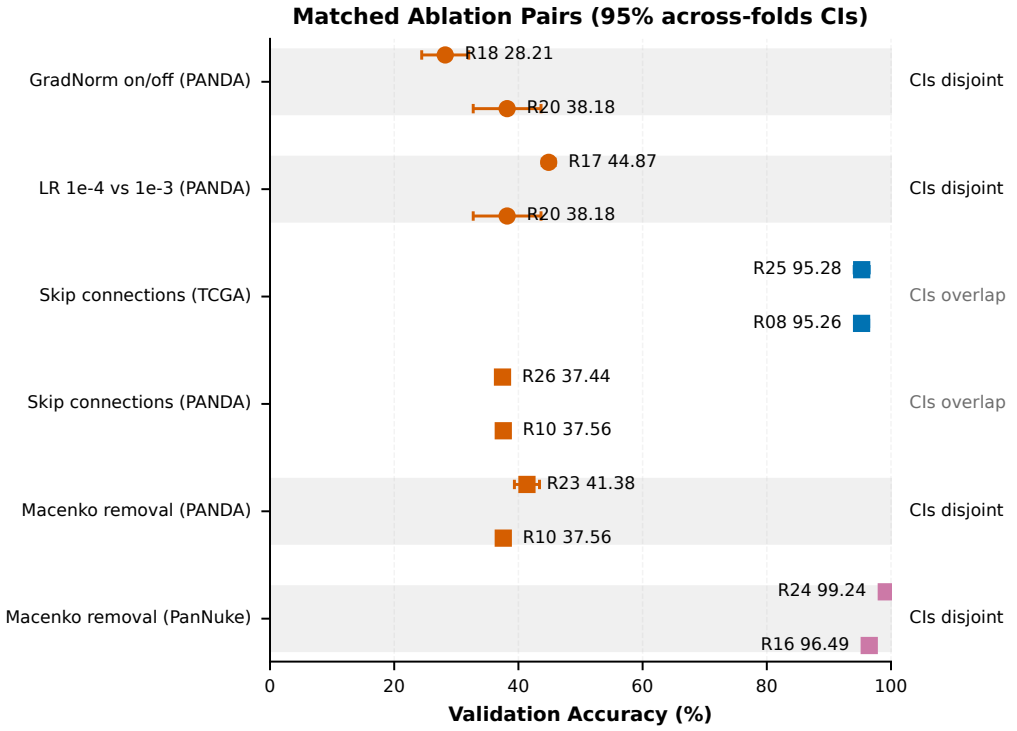


Figure 7: Matched-ablation forest plot of validation accuracy with 95% bootstrap CIs across the five folds for the six strictly matched pairs. Shaded rows highlight pairs with disjoint marginal CIs: isolated GradNorm (Run 18, 28.21%) vs. static 5:1 (Run 20, 38.18%); LR 10^{-4} vs. 10^{-3} (Runs 17 vs. 20: 44.87% vs. 38.18%); no-Macenko PANDA (Run 23, 41.38%) vs. matched (Run 10, 37.56%); no-Macenko PanNuke (Run 24, 99.24%) vs. matched (Run 16, 96.49%). Unshaded rows indicate pairs with overlapping marginal CIs: skip ablation TCGA (Runs 25 vs. 08: 95.28% vs. 95.26%, $\Delta = +0.02$); skip ablation PANDA (Runs 26 vs. 10: 37.44% vs. 37.56%, $\Delta = -0.12$). The right-margin status column descriptively categorizes CI overlap and separation across folds; it is not a formal test of paired equivalence.

71.32], $\Delta = +6.97$). Normalization altered the image color representation and was associated with lower classification accuracy in these matched experiments; the exact histological mechanism remains undetermined. Figure 7 consolidates all six matched pairs; Figure 3 visualizes the chromatic contrast alteration.

5.3. Isolating Learning Rate from Dynamic Gradient Balancing

Group 4 (PANDA × VGG16) factorizes the key optimization hyperparameters: learning rate ($\eta \in \{10^{-3}, 10^{-4}\}$), GradNorm-style balancing (on/off), and static loss-weighting ratios $\lambda_{seg} : \lambda_{cls} \in \{1{:}10, 1{:}1, 5{:}1, 10{:}1\}$, all under raw inputs.

Direct Isolation of Dynamic Gradient Balancing (Run 18).. The isolated GradNorm arm at $\eta = 10^{-3}$ under matched raw-input conditions (Run 18: PANDA × VGG16, GradNorm $\alpha = 1.5$, raw input) compares directly against the static 5:1 baseline (Run 20). Under these matched conditions, activating GradNorm ($\alpha = 1.5$) resulted in a validation accuracy decrease from 38.18% to 28.21% (95% bootstrap CI across the five folds: $[24.76, 31.47]$ vs $[32.75, 41.79]$, $\Delta_{\text{Acc}} = -9.97$ percentage points), while fold validation macro-Dice remained identical at 84.39%.

This 28.21% accuracy mirrors the lower accuracy observed in Phase v2 at $\eta = 10^{-4}$ (Run 09: 33.34% Acc). We hypothesize that dynamic gradient balancing in this configuration induces destructive gradient interference on the shared encoder, consistent with observations in multi-task optimization studies [19, 14]. However, gradient directional alignment was not directly measured during training, and isolating whether the decrease stems specifically from the $\alpha = 1.5$ asymmetry parameter would require a parametric sweep over α under identical matched conditions.

Learning Rate and Loss Weighting Dynamics.. Holding GradNorm off, reducing the learning rate from $\eta = 10^{-3}$ (Run 20: 38.18% Acc) to $\eta = 10^{-4}$ (Run 17: 44.87%) yields a statistically resolvable improvement, with disjoint 95% bootstrap CIs across the five folds. The static loss-ratio sweep (Runs 19–22) confirms $\lambda = 5{:}1$ and 10:1 as comparable static balances (36.51%–39.71%), yet no ratio approaches the claimed $87\% - 88\%$ (Figure 8). The isolated GradNorm arm (Run 18) drops to 28.21% against the 38.18% static baseline with non-overlapping CIs, and the canonical decoupled scheme plateaus at 22.70% (95% bootstrap CI $[14.66, 30.73]$). Dynamic balancing in either formulation cannot bridge the gap (Figures 6, 7).

5.4. Limitations

Several methodological limitations of this audit should be explicitly considered:

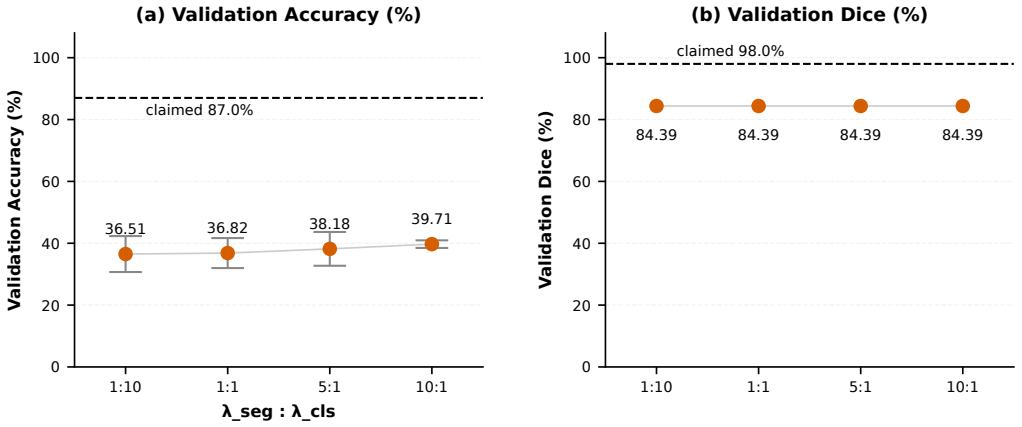


Figure 8: Static loss-weight ratio sweep on PANDA × VGG16 (raw input, $\eta = 10^{-3}$, no GradNorm). (a) Validation accuracy with 95% bootstrap-CI whiskers across the five folds: $\lambda_{seg} : \lambda_{cls}$ = 1:10 (36.51%), 1:1 (36.82%), 5:1 (38.18%), 10:1 (39.71%); the black dashed line marks the claimed 87.0% accuracy. (b) Validation Dice for the same four arms (84.39%, 84.39%, 84.39%, 84.39%) against the claimed 98.0%. The ratios yield comparable accuracy (36.51% – 39.71%), yet no ratio approaches the claimed target — the gap is not a loss-weighting artifact.

1. *Reconstruction of Underspecified Protocols:* Because the baseline authors released no training code, seed policies, or split manifests, our reproduction reconstructed the pipeline faithfully from published text descriptions and official benchmark repositories. Where published protocols could not be reconstructed due to unaligned benchmark releases (such as the discrepancy between ISIC 2018 Task 1 with 2,594 segmentation masks and Task 3 with 10,015 classification labels without cohort split manifests), we evaluated the established multi-tissue PanNuke benchmark instead.
2. *Thumbnail vs. Whole-Slide Multiscale Analysis:* Our audit directly tested standard 2D multi-task U-Nets on fixed-resolution tiles as specified in the published recipe. While necessary to evaluate the claims directly, thumbnail-level classification cannot capture the multiscale context required for gigapixel whole-slide grading in computational pathology.
3. *Computational Budget and Seed Scope:* Each of the 27 configurations was evaluated across 5-fold cross-validation under fixed seeds per fold (135 total fold-runs) with an early-stopping patience of 10

epochs. While fold-level bootstrap confidence intervals characterize data-partitioning uncertainty, evaluating multiple random seeds per fold across all 135 models was computationally prohibitive.

4. *Metric Estimator Sensitivity:* As demonstrated in Section 3.2, biomedical segmentation metrics depend heavily on aggregation conventions. We disclosed both batch-pooled validation macro-Dice and per-case empty-credit Dice to make these dependencies transparent.

6. Conclusion & Recommendations

Across 27 configurations $\times$ 5 folds (135 fold-cells), reported near-saturated metrics (88% – 99%) did not reproduce under rigorous cross-validation. The performance gap stems from dataset complexity, task-formulation divergence, and multi-task optimization trade-offs: lateral skip connections exhibit modest paired differences (+0.02 to −0.12 pts), while dynamic gradient balancing ($\alpha = 1.5$) is associated with lower validation accuracy (38.18% $\rightarrow$ 28.21% Acc). We recommend: (1) mandatory reporting of sample and patient grouping cardinalities, ensuring independent partition units; (2) explicit mathematical definition of the Dice evaluation convention, reporting foreground-only lesion overlap on sparse datasets alongside negative-case false-positive rates; (3) reporting 95% bootstrap confidence intervals and descriptive paired comparisons on matched ablations to distinguish architectural effects from fold variance; (4) validating pathology models on aligned clinical tasks and multiscale benchmarks rather than low-resolution proxy thumbnails; and (5) systematic evaluation of multi-task balancing hyperparameters before clinical translation.

None of the mechanisms quantified here is exotic. Each is grounded in the evaluation literature (Section 2), and the study whose figures motivated this audit is already being cited as evidence of multi-task efficacy in follow-on oncology work. The recommendations above are offered as a verification checklist for multi-task detection and segmentation models intended for clinical consumption.

Declaration of generative AI and AI-assisted technologies in the writing process

AI-assisted tools were used for language refinement and figure preparation. All scientific content, experiments, data analysis, and conclusions are the sole work of the authors, who reviewed and approved the final manuscript.

CRediT author statement

Muhammad Abdullah Ali: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Project administration. Muhammad Ibrahim Kiani: Writing – review & editing. Muhammad Abdullah Aamir: Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All training scripts, per-run JSON summaries, evaluation checkpoints, generated figures, and the complete results matrix with 95% bootstrap confidence intervals (`paper_results_matrix_with_ci.csv`) are publicly available at https://github.com/ApatheticMioz/cancer-pathology-dl.

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